IMU-Based Gait and Heart Rate Analysis in Horses Using EquiPro and Polar V800

Introduction

Lameness, an alteration of the normal gait due to a functional or structural disorder, commonly attributed to orthopedic pain in horses [1], is a prevalent issue that often results in costly health complications. Veterinarians typically rely on visual examinations to assess lameness, but these methods are subjective and come with many limitations. This project aims to study the walk, trot, and canter gaits in several horses of different breeds using wireless inertial measurement units (IMUs) of the EquiMoves system, which provide an objective means of identifying lameness and monitoring rehabilitation. Observed gait asymmetries due to past injuries can serve as indicators of continued pain or discomfort. The IMUs capture detailed horse movement data, and the EquiMoves software generates an in-depth analysis of each gait. Access to the semi-processed accelerometer data facilitates mathematical analysis in our research setting. In addition, a Polar V800 heart rate monitor was briefly used to collect inter-beat interval data, offering a non-invasive method to assess physiological responses. By synchronizing data from both systems, either by time, by stride, or through another appropriate method, we can explore how gait type affects asymmetry levels and whether heart rate data can support the detection of instantaneous moments of pain during movement.

Methods

The data was collected by lunging the horses in an indoor arena at the walk, trot, and canter, in both clockwise and counter-clockwise directions. Seven time-synchronized wireless inertial measurement units (IMUs), each with a precision of 100 ns, were placed on all four limbs, the poll, the withers, and the sacrum. These IMUs transmitted data to an Inertia Gateway receiver, and their embedded gyroscopes and accelerometers were used to determine the 3D orientation of each sensor. In a few of the later sessions, a Polar V800 heart rate monitor was also used to collect inter-beat interval data. The monitor consisted of a chest strap and receiver that allowed non-invasive, continuous heart rate tracking during lunging. To enable meaningful comparison and analysis, all IMU and heart rate data must be synchronized as closely as possible, either by timestamp alignment, gait cycle, or stride segmentation, before further processing and analysis.

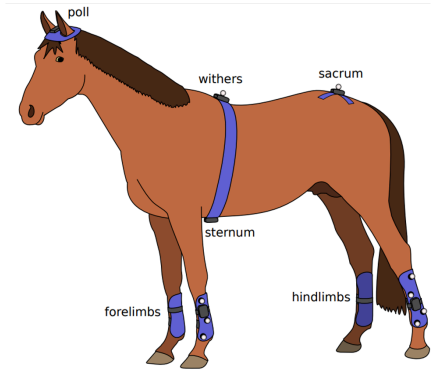

Figure 1: Placement of EquiPro IMUs on horse. [1]

Results

The Root Mean Square of Successive Differences (RMSSD) is a widely used time-domain measure of heart rate variability that captures short-term fluctuations in the intervals between successive heartbeats, or RR intervals. It is primarily associated with parasympathetic nervous system activity, reflecting a “rest-and-digest” state. Higher RMSSD values indicate greater beat-to-beat variability, typically linked to relaxation and recovery, while lower values may suggest stress, fatigue, or discomfort. In Figure 2, all six horses exhibited higher RMSSD values when trotting to the left during the recorded session, indicating improved parasympathetic tone in that direction. This trend is consistent with the common handling practice of leading horses from the left side, which may make leftward movement more familiar and less stressful.

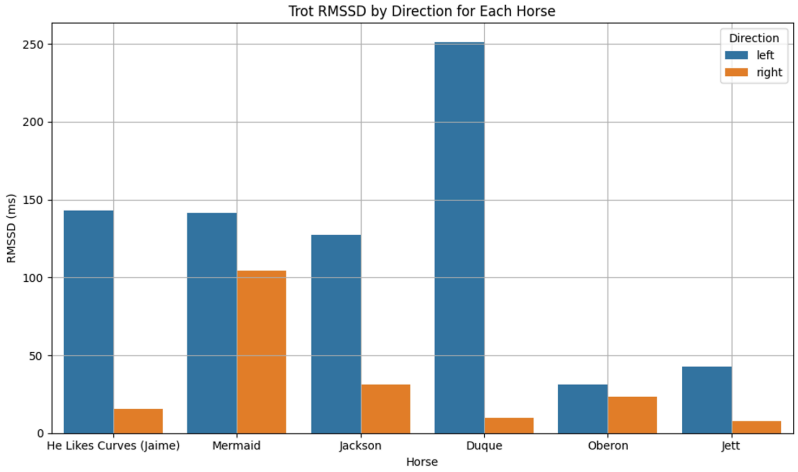

Figure 2: Root Mean Square of Successive Differences between RR intervals for each horse trotting left and right during one session.

Figure 3 shows the six horses ranked by relative recovery rate, calculated from two heart rate measurements (HR1 and HR2) taken during the transition between lunge directions. HR1 is measured at the end of Gait Direction 1, and HR2 at the end of the transition period before Gait Direction 2 begins. Using the resting HR measured before lunging as the baseline, the relative recovery rate is defined as:

$$Relative\ Recovery\ (\%) = \frac{HR1 - HR2}{HR1 - Resting\ HR} \times 100.$$

The numerator represents the actual heart rate drop during the transition, while the denominator is the maximum possible drop if the horse had fully recovered to resting heart rate. As expected, younger and more agile horses show higher recovery rates between lunging directions.

Horse	Recovery (%)
Oberon	68.80%
Jaime	59.17%
Jett	49.72%
Jackson	29.19%
Mermaid	17.45%
Duque	11.41%

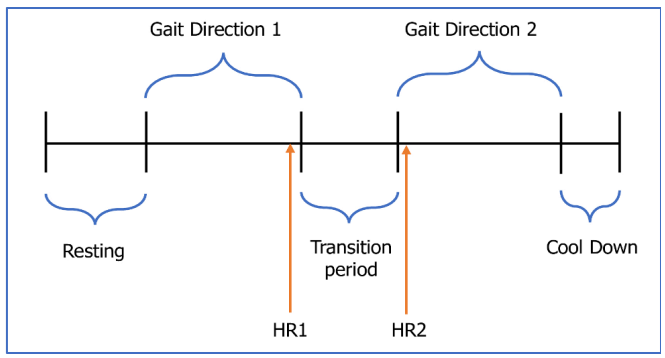

Figure 3: Table with relative recovery rate of each horse and diagram of lunging session with metrics used to calculate recovery.

Future Directions

Future work will focus on improving data collection consistency and expanding the scope of analysis. When measuring RMSSD for both left and right directions, it will be beneficial to lunge each horse for longer periods, approximately five minutes in both clockwise and counter-clockwise directions, to ensure sufficient heart rate data. We will also alternate the initial lunging direction between sessions to avoid always starting to the left, the more familiar and comfortable direction for most horses. Because current results are based on a single session per horse, future analyses will incorporate data from multiple sessions and from all gait types, rather than only the trot, allowing us to calculate more representative average RMSSD values.

Our immediate priority is to establish a streamlined workflow for exporting data after each lunging session. Since the EquiMoves and Polar systems operate independently, a key

challenge is synchronizing their outputs in real time to generate a unified data matrix, with each row representing the combined measurements at a single moment in time. We are also refining methods to extract and align continuous accelerometer metrics across time. Long-term, we aim to develop a dyadic, integrated model that captures the mutual interactions between horse and human.

References

1. Harder, V., Serra Bragança, F. M., Rhodin, M., Voskamp, J. P., Van Weeren, P. R., Roepstorff, L., & Barneveld, A. (2019). *EquiMoves: A wireless networked inertial measurement system for objective examination of horse gait*. Sensors, 19(2), 480. https://doi.org/10.3390/s19020480